

2023] (8)

$$Z(T, V, N) = \frac{V^N}{N!} \left(\frac{2\pi m k T}{h^2} \right)^{3N/2} Z_{int}(T)^N$$

i) $PV = NkT$ *

• $F = -kT \log(Z)$

• $P = - \left(\frac{\partial F}{\partial V} \right)_{N, T}$

$\log(x^n) = n \log(x)$

$$F = -kT \log(Z) = -kT \left[\log(V^N) + \log \left(\frac{(2\pi m k T)^{3N/2}}{N! h^2} Z_{int}(T) \right) \right]$$

$$P = - \left(\frac{\partial F}{\partial V} \right)_{N, T} = +kTN \frac{\partial}{\partial V} (\log(V))$$

$$= \frac{kTN}{V}$$

$$\log_x(y) = \frac{\log_r(y)}{\log_r(x)}$$

$$PV = NkT$$

ii) Entropy $S = - \left(\frac{\partial F}{\partial T} \right)_{V, N}$ | $S = \frac{U - F}{T}$ *

* $U = ST + F$ *

$$F = -kT \left[\log \left(T^{3N/2} Z_{int}(T)^N \right) + \log \left(\frac{V^N}{N!} \left(\frac{2\pi m k}{h^2} \right)^{3N/2} \right) \right]$$

$$\begin{aligned} \left(\frac{\partial \log}{\partial T} \right)_{V, N} &= -kT \frac{\partial}{\partial T} \left(\log(T^{3N/2} \cdot Z_{int}(T)^N) \right) \\ &= -kT \frac{\partial}{\partial T} \left(\frac{3N}{2} \log(T) + N \log(Z_{int}(T)) \right) \\ &= -kT \left(\frac{3N}{2} \frac{1}{T} + \frac{N}{Z_{int}(T)} \frac{\partial Z_{int}}{\partial T} \right) \\ &= kTN \left(\frac{3}{2} \frac{1}{T} + \frac{1}{Z_{int}(T)} \frac{\partial Z_{int}}{\partial T} \right) \end{aligned}$$

Also have

$$- \frac{\partial}{\partial T} (kT) [\log(\quad)] = -k \log[\quad]$$

$$\left(\frac{\partial F}{\partial T}\right)_{N,T} = -k \log(z) - kTN \left(\frac{3}{2} \frac{1}{T} + \frac{1}{z_{int}} \frac{\partial z_{int}}{\partial T} \right)$$

$$F = -kT \log(z) \quad \rightarrow \quad \frac{F}{T}$$

$$S = - \left(\frac{\partial F}{\partial T} \right)_{N,T} = \frac{F}{T} + kTN \left(\frac{3}{2} \frac{1}{T} + \frac{1}{z_{int}} \frac{\partial z_{int}}{\partial T} \right)$$

$$U = ST + F$$

$$U = -F + kT^2 N \left(\frac{3}{2} \frac{1}{T} + \frac{1}{z_{int}} \frac{\partial z_{int}}{\partial T} \right) + F$$

$$U = kTN \left(\frac{3}{2} + \frac{1}{z_{int}} \frac{\partial z_{int}}{\partial T} \right)$$

$$z_{int} = 1$$

→ modification to energy by z_{int}
 ~D from other non-ideal forces
 that we introduce.

$$U = \left(\frac{3}{2} kT \right) N$$

↑ thermal
 (average)

• electric, ...

NOT potential

